

Adversarial IR

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What Adversarial IR Is – Intro

Adversarial Information Retrieval - a topic in information retrieval related to strategies for working with a data source where some portion of it has been manipulated maliciously.

What exactly does Adversarial involve?

Gathering, indexing, filtering, retrieving and ranking information from data that has been manipulated maliciously.

Adversarial IR includes the study of methods to detect, isolate, and defeat such manipulation.

The 4 major types of attacks

The key types of adversarial IR designed to manipulate Machine Learning models include:

1

Poisoning

The corruption of training data

2

Evasion

Messing with predictions

3

Extraction

Model/data information stealing

4

Inference

Leaking sensitive data

Poisoning

Poisoning attacks occur when attackers slip malicious crafted data into the training dataset to shift how the model makes decisions. This skews how the model learns and performs, leads to bias outputs and degraded accuracy.

These attacks are categorized into two subtypes:

- Non-targeted - Forces the model to produce any incorrect outputs, aimed to degrade the overall performance
- Targeted - Forces the model to produce specific and often predefined misclassified outputs

Evasion

Evasion attacks occur by deceiving already trained machine learning models. It does this by making small changes until the model's answer is misclassified

Examples of evasion attacks:

- Fast gradient sign method (FGSM) - nudge inputs in directions that will most confuse the model
- Projected Gradient Descent (PGD) - repeated version of FGSM that keeps changing the input until the model gets it wrong

Extraction

Extraction occurs when someone copies a trained machine learning model by repeatedly querying, recording the outputs, and then using it to train their copy of the model ('black-box' view). This then gives the attackers a 'white-box' view to easily employ evasion attacks

- White-box: attacker has complete access to the targets model (including its architecture and parameters)
- Black-box: attacker has no knowledge of internal workings and rely on observing outputs to achieve their goals

Inference

Inference attacks expose confidential data without breaching backend systems from language models. This is done by probing a model's response to uncover hidden aspects of a model's internal structure using model inversion.

- Query based attacks - Output of crafted queries is used to infer the parameters or architecture of said model
- Membership inference attacks - Determines if a data point was used to train by model by querying the data point and analyzing response

How we can defend from attacks

Defend against evasion attacks

Enhancing model robustness through adversarial training and implementing input validation such as region-based classification and the analysis of nearby data points, instead of only one input.

Defend against poisoning attacks

Cleaning and validating training data, detecting unusual patterns, and using robust training methods. Advanced systems like RECESS can identify malicious participants during federated learning.

Defend against inference attacks

Reducing overfitting, using differential privacy, and applying methods like RelaxLoss, which makes training and non-training data appear more similar to attackers.

Defend against extraction attacks

Rate limiting model queries and carefully monitoring for suspicious activity, and watermarking outputs. Advanced methods like MisGUIDE disrupt malicious extraction attempts.

Real-World Systems It Applies To

2011 JCPenny - Spam / Web Spam

Wikipedia or Reddit Poisoning Attack

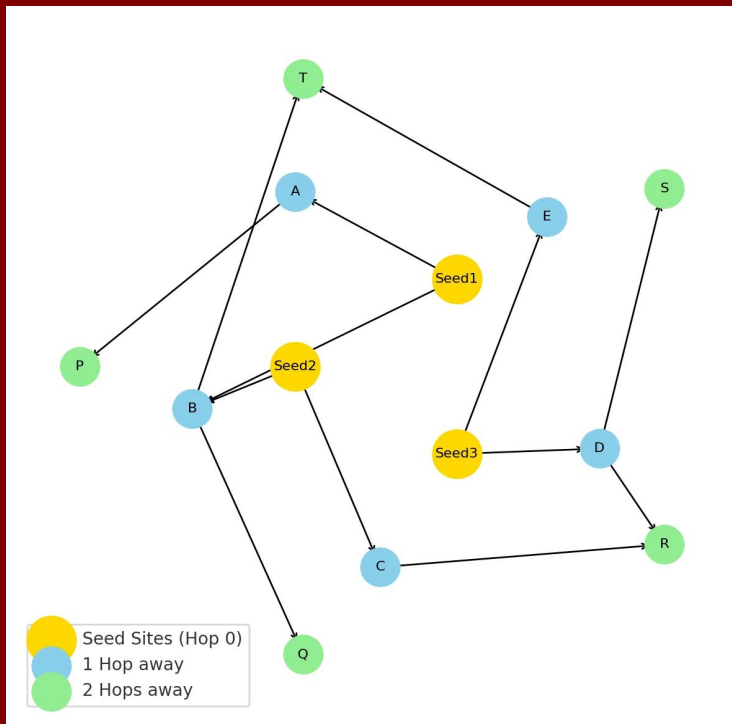
Proprietary ML models with standard API access

How to stop malicious ranking attacks

- Spam Rank or Trust Rank - collecting a group of trusted web pages in order to penalize spam links
- Adversarial Training for Neural Rankers - training BERT-based rankers on augmented corpora that includes spam examples so the model learns to ignore them

Prediction of Spam

- Number of Words in a Page < Number of Words in the Title of a Page
- Fraction of Visible Content on Page
- Number of Advertisements on Page
- Too Many Call-to-Action Phrases (Hurr now, Limited Offer, only x items left)
- Low Count of Stop Words and Low Image Count



Trust Rank

Google Spambrain

Large-Scale spam classifiers like google's spam-detection algorithm is designed to identify spam and low-quality websites with high accuracy. The algorithms "learn" from your website, and it improves over time. If it finds that your site is still spammy, it'll continue to punish you.

Limitations of Adversarial Defenses

Collateral damage - some spam filters can be very aggressive and penalize legitimate pages. This can create some unintended bias towards more established domains.

LLM-generated content - as LLM get more advanced, they can produce more reputable looking and coherent text for websites. So low-quality websites or malicious content is harder to detect

RAG Poisoning - RAG systems can sometimes include adversarially crafted documents leading to a hijacked model response.

Links to research

- [Arxiv.org](https://arxiv.org) - A Survey on Adversarial Information Retrieval on the Web
- [Lumenova.ai](https://lumenova.ai) - Types of Adversarial Attacks and How To Overcome Them
- [Sentinelone.com](https://sentinelone.com) - What Are Adversarial Attacks? Threats & Defenses
- [Knostic.ai](https://knostic.ai) - Adversarial AI Attacks & How to Stop Them
- [Nightfall.ai](https://nightfall.ai) - AI Security Model Inversion
- [Specterops.io](https://specterops.io) - Learning Machine Learning Part 3: Attacking Black Box Models
- [Paloaltonetworks.com](https://paloaltonetworks.com) - What Is an Adversarial AI Attack?